

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings

Code:-

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

/* Global variable */
int global_var = 100;

/* Global uninitialized variable */
int global_uninitialized;

/* Static global variable */
static int static_global = 200;

/* Function - stored in code/text segment */
void show_addresses(void)
{
    /* Static local variable */
    static int static_local = 300;

    /* Automatic local variable - stack */
    int stack_var = 400;

    /* Heap variable */
    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL) {
        perror("malloc");
        exit(EXIT_FAILURE);
    }

    *heap_var = 500;

    printf("Process ID          : %d\n", getpid());
    printf("-----\n");

    printf("Code / Function address : %p\n", (void *)show_addresses);
```

```
printf("Global variable address : %p\n", (void *)&global_var);

printf("Global uninitialized   : %p\n",
      (void *)&global_uninitialized);

printf("Static global address  : %p\n",
      (void *)&static_global);

printf("Static local address   : %p\n",
      (void *)&static_local);

printf("Stack variable address : %p\n",
      (void *)&stack_var);

printf("Heap variable address  : %p\n",
      (void *)heap_var);

printf("-----\n");

printf("Press Enter to terminate...\n");
getchar();

free(heap_var);
}

int main(void)
{
    show_addresses();
    return 0;
}
```

Output:

Terminal 1

```
adithya@LAPTOP-07A78KDI:~/OS$ nano memory_layout.c
adithya@LAPTOP-07A78KDI:~/OS$ gcc -Wall -Wextra -O0 -g memory_layout.c -o memory_layout
adithya@LAPTOP-07A78KDI:~/OS$ ./memory_layout
Process ID          : 573
-----
Code / Function address : 0x579f1cacb249
Global variable address : 0x579f1cace010
Global uninitialized   : 0x579f1cace020
Static global address  : 0x579f1cace014
Static local address   : 0x579f1cace018
Stack variable address : 0x7ffd27552efc
Heap variable address  : 0x579f529d0010
-----
Press Enter to terminate...

adithya@LAPTOP-07A78KDI:~/OS$ cat /proc/12345/maps
cat: /proc/12345/maps: No such file or directory
adithya@LAPTOP-07A78KDI:~/OS$ ./memory_layout
Process ID          : 721
-----
Code / Function address : 0x5b032f1e2249
Global variable address : 0x5b032f1e5010
Global uninitialized   : 0x5b032f1e5020
Static global address  : 0x5b032f1e5014
Static local address   : 0x5b032f1e5018
Stack variable address : 0x7ffe1d760ecc
Heap variable address  : 0x5b0335022010
-----
Press Enter to terminate...
```

Terminal 2

```
adithya@LAPTOP-07A78KDI:~$ cd OS
adithya@LAPTOP-07A78KDI:~/OS$ cat /proc/12345/maps
cat: /proc/12345/maps: No such file or directory
adithya@LAPTOP-07A78KDI:~/OS$ cat /proc/721/maps
5b032f1e1000-5b032f1e2000 r--p 00000000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e2000-5b032f1e3000 r-xp 00001000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e3000-5b032f1e4000 r--p 00002000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e4000-5b032f1e5000 r--p 00002000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e5000-5b032f1e6000 rw-p 00003000 08:30 15918 /home/adithya/OS/memory_layout
5b0335022000-5b0335043000 rw-p 00000000 00:00 0 [heap]
744a19e00000-744a19e28000 r--p 00000000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a19e28000-744a19fc0000 r-xp 00028000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a19fc0000-744a1a00e000 r--p 001c0000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a1a00e000-744a1a012000 r--p 0020d000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a1a012000-744a1a014000 rw-p 00211000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a1a014000-744a1a021000 rw-p 00000000 00:00 0
744a1a124000-744a1a127000 rw-p 00000000 00:00 0
744a1a12f000-744a1a131000 rw-p 00000000 00:00 0
744a1a131000-744a1a135000 r--p 00000000 00:00 0 [vvar]
744a1a135000-744a1a137000 r--p 00000000 00:00 0 [vvar_vclock]
744a1a137000-744a1a139000 r-xp 00000000 00:00 0 [vdso]
744a1a139000-744a1a13a000 r--p 00000000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a13a000-744a1a169000 r-xp 00001000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a169000-744a1a174000 r--p 00030000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a174000-744a1a176000 r--p 0003b000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a176000-744a1a177000 rw-p 0003d000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a177000-744a1a178000 rw-p 00000000 00:00 0
7ffe1d742000-7ffe1d763000 rw-p 00000000 00:00 0 [stack]
adithya@LAPTOP-07A78KDI:~/OS$ grep '\[heap\]' /proc/721/maps
5b0335022000-5b0335043000 rw-p 00000000 00:00 0 [heap]
adithya@LAPTOP-07A78KDI:~/OS$ grep '\[stack\]' /proc/721/maps
7ffe1d742000-7ffe1d763000 rw-p 00000000 00:00 0 [stack]
adithya@LAPTOP-07A78KDI:~/OS$ grep memory_layout /proc/721/maps
5b032f1e1000-5b032f1e2000 r--p 00000000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e2000-5b032f1e3000 r-xp 00001000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e3000-5b032f1e4000 r--p 00002000 08:30 15918 /home/adithya/OS/memory_layout
```

```

adithya@LAPTOP-67A78KDI:~/OS$ ps -x 721
721:  ./memory_layout
Address      Kbytes      RSS      Dirty Mode      Mapping
00005b032f1e1000      4      4      0 r----- memory_layout
00005b032f1e2000      4      4      0 r-x-- memory_layout
00005b032f1e3000      4      4      0 r----- memory_layout
00005b032f1e4000      4      4      4 r----- memory_layout
00005b032f1e5000      4      4      4 rw---- memory_layout
00005b0335022000     132      4      4 rw---- [ anon ]
0000744a19e00000     160     156      0 r----- libc.so.6
0000744a19e28000    1632    1120      0 r-x-- libc.so.6
0000744a19fc0000     312     184      0 r----- libc.so.6
0000744a1a00e000      16      16     16 r----- libc.so.6
0000744a1a012000       8       8       8 rw---- libc.so.6
0000744a1a014000      52      20     20 rw---- [ anon ]
0000744a1a124000      12       8       8 rw---- [ anon ]
0000744a1a12f000       8       4       4 rw---- [ anon ]
0000744a1a131000      16       0       0 r----- [ anon ]
0000744a1a135000       8       0       0 r----- [ anon ]
0000744a1a137000       8       8       0 r-x-- [ anon ]
0000744a1a139000       4       4       0 r----- ld-linux-x86-64.so.2
0000744a1a13a000     188     184      0 r-x-- ld-linux-x86-64.so.2
0000744a1a169000      44      44       0 r----- ld-linux-x86-64.so.2
0000744a1a174000       8       8       8 r----- ld-linux-x86-64.so.2
0000744a1a176000       4       4       4 rw---- ld-linux-x86-64.so.2
0000744a1a177000       4       4       4 rw---- [ anon ]
00007ffe1d742000     132      16     16 rw---- [ stack ]

total kB      2768     1812     100
adithya@LAPTOP-67A78KDI:~/OS$ cat /proc/721/maps
5b032f1e1000-5b032f1e2000 r--p 00000000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e2000-5b032f1e3000 r-xp 00001000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e3000-5b032f1e4000 r--p 00002000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e4000-5b032f1e5000 r--p 00002000 08:30 15918 /home/adithya/OS/memory_layout
5b032f1e5000-5b032f1e6000 rw-p 00003000 08:30 15918 /home/adithya/OS/memory_layout
5b0335022000-5b0335043000 rw-p 00000000 00:00 0 [heap]
744a19e00000-744a19e28000 r--p 00000000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a19e28000-744a19fc0000 r-xp 00028000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a19fc0000-744a1a00e000 r--p 001c0000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a1a00e000-744a1a012000 r--p 00280000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a1a012000-744a1a014000 rw-p 00211000 08:30 13645 /usr/lib/x86_64-linux-gnu/libc.so.6
744a1a014000-744a1a021000 rw-p 00000000 00:00 0
744a1a124000-744a1a127000 rw-p 00000000 00:00 0
744a1a12f000-744a1a131000 rw-p 00000000 00:00 0
744a1a131000-744a1a135000 r--p 00000000 00:00 0 [vvar]
744a1a135000-744a1a137000 r--p 00000000 00:00 0 [vvar_vclock]
744a1a137000-744a1a139000 r-xp 00000000 00:00 0 [vdso]
744a1a139000-744a1a13a000 r--p 00000000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a13a000-744a1a169000 r-xp 00001000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a169000-744a1a174000 r--p 00030000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a174000-744a1a176000 r--p 0003b000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a176000-744a1a177000 rw-p 0003d000 08:30 13642 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
744a1a177000-744a1a178000 rw-p 00000000 00:00 0
7ffe1d742000-7ffe1d763000 rw-p 00000000 00:00 0 [stack]
adithya@LAPTOP-67A78KDI:~/OS$

```